

CS2300: File Structures and Introduction to Database Systems

Lecture 19: (Ch 10) SQL Programming Techniques

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Outline

- Accessing databases from programs
- Host languages
- Data sublanguages
- Database programming languages



```
ConnectionString="Database=09_home; User=sa; Password=1234567890"
DBProvider = "Database provider"
SelectSQL1 = " Select id, name, quantity from all"
QuerySQL1 = " where id between 10000000 and 10000000"
QuerySQL2 = " group by id, name"
SelectQuery = SelectSQL1 & QuerySQL1 & QuerySQL2
Execute Query; Commit Transaction; Select new data
Form Navigation
If KeyAscii = 13 Then Execute Query
If Not Chr(KeyAscii) Like "#* And KeyAscii < 9 Then
```

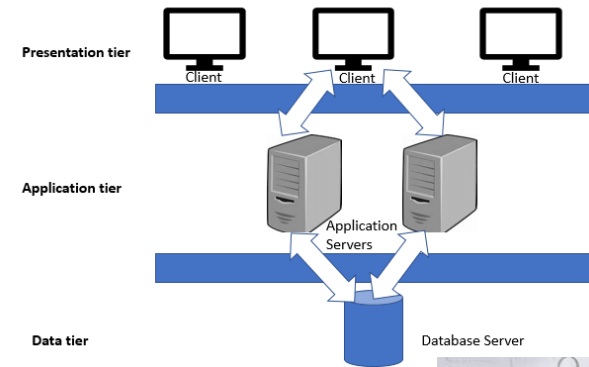
SQL in Real Programs

- We have seen only how SQL is used at the generic query interface --- an environment where we sit at a terminal and ask queries of a database.
 - Convenient for schema and constraint creation
 - Occasional ad hoc queries
- Reality is almost always different: conventional programs interacting with SQL
 - Application programs/database applications
 - Database access via web interface

Typical Sequence of Interaction in Database Programming

Three-tier client/server model

- A common architecture for database access
- Client program
 - Handles display of information
- Application program
 - A middle-tier that implements the logic of a business software application
- Database servers
 - For accessing or update the data

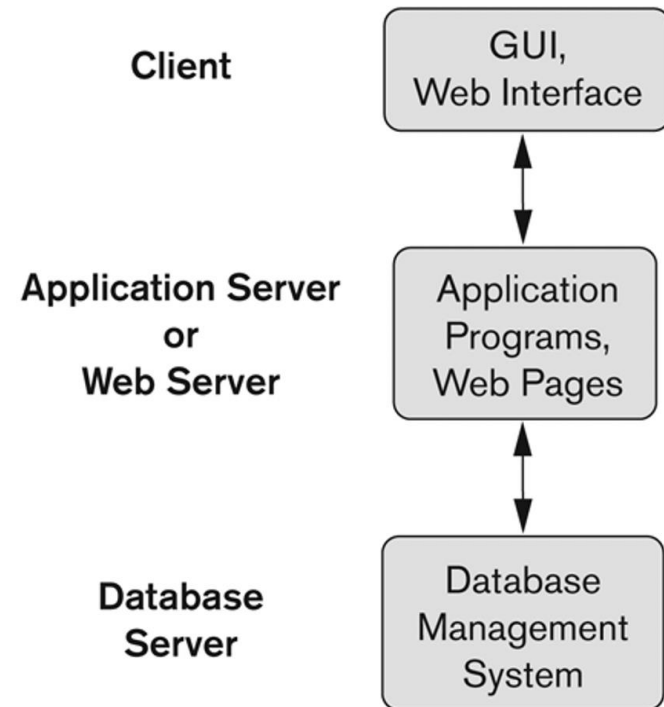


1. When the application program requires access to a database, first must establish or **open** a connection to the database server.
2. Once the connection is established, the program can then **interact** with the database by submitting queries, updates, and other database commands
3. When the program no longer needs access to a particular database, it should terminate or **close** the connection.

Three-Tier Architectures for Web Applications

three-tier architecture:

- Common for many web applications
- An intermediate layer is added between the client and the database server
- *What might the main role of the middle layer be for a web application?*



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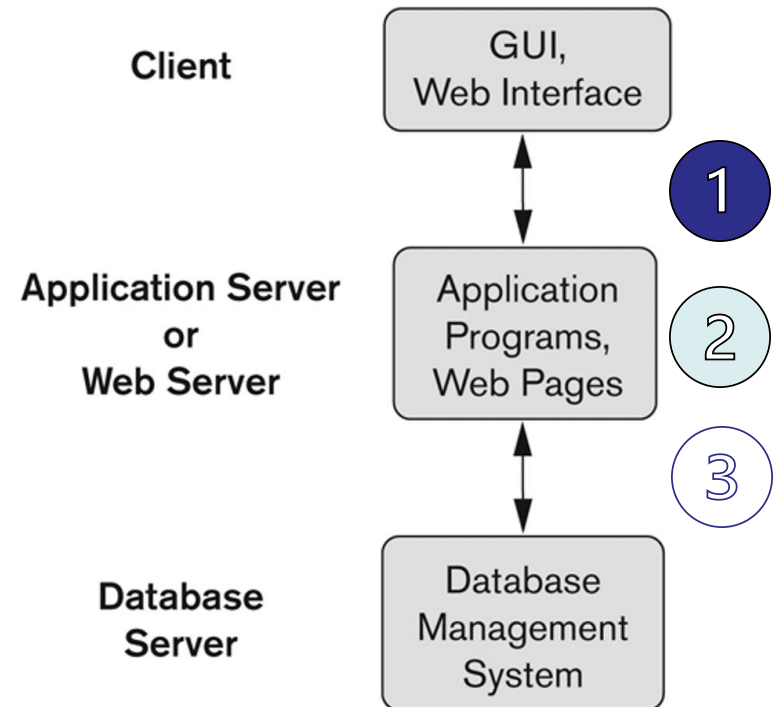
the middle tier:

- Serve the role of an application server or a web server
- Main responsibility:
 - *Running the application programs*
 - *Storing business rules (i.e. procedures or constraints)*
 - *Improving database security*
 - *e.g. checking a client's credentials before allowing access to the database server*

Can you name an example of a web app that potentially uses the 3-tier architecture?

Three-Tier Architectures for Web Applications

- (1) Accepts **requests** from the client (*the **user interface***)
- (2) Processes the requests and prepare **database queries** and commands to the ?_____? server
 - (1) Based on **application rules**
- (3) Acts as a ?_____? for passing (partially) **processed data** from the database server back to the clients
 - (1) Support **data access** management
 - (2) The client layer may process and ?_____? the data further to be **presented** to the users



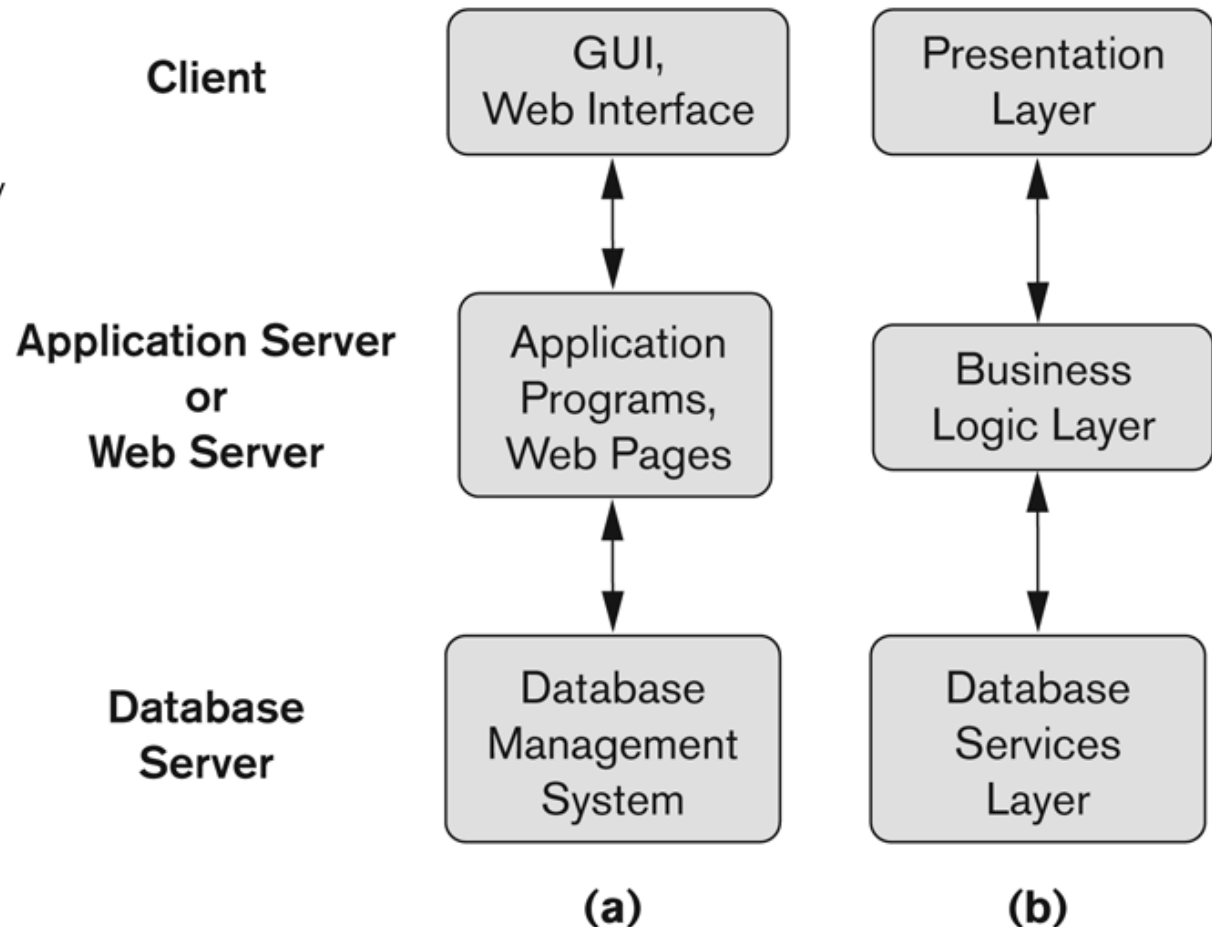
Three-tier client-server architecture

Figure 2.7

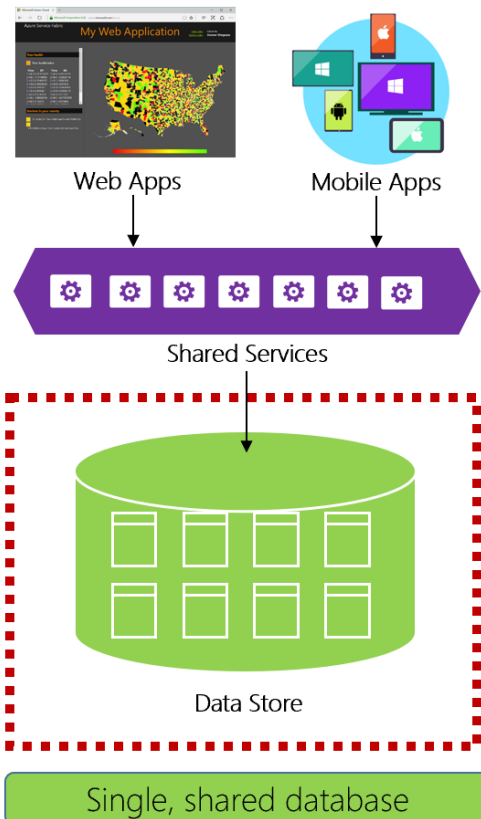
Logical three-tier client/server architecture, with a couple of commonly used nomenclatures.

The middle layer can also act as a web server

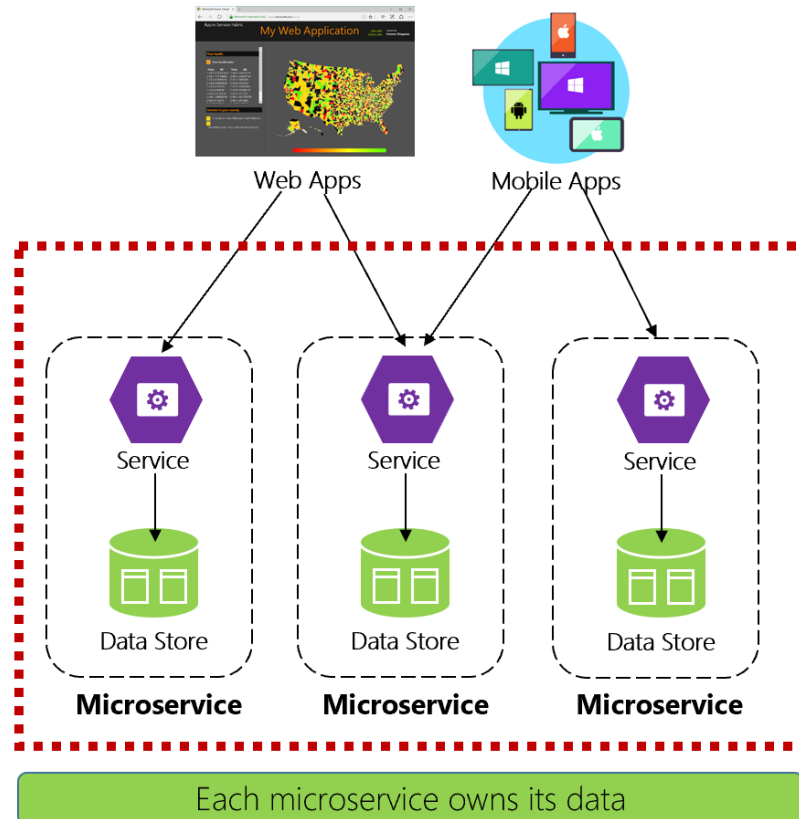
- Retrieves **query results** from the db server
- **Formats** them into dynamic web pages
- Pages are then **viewed** by the web browsers at the client side
 - client machine: typically a PC. Or mobile device



Monolithic Application



Cloud-Native Application



Approaches to Database Programming

1. Using a library of database functions or classes (*Application programming interface - API*)

- Utilizing a set of pre-written functions or classes provided by an API to interact with a database from a host programming language
- The development of a database API is typically the responsibility of the database system's developers.
- *Example:* Python with SQLite (a lightweight disk-based, serverless database)
 - `sqlite3` built-in library
 - [Code examples](#)

Which approaches are the more common?

Approaches to Database Programming

2. Embedding database commands in a general-purpose programming language

- Embeds database commands *directly* into the code of the host programming languages
- Typically requires a precompiler/preprocessor to convert these embedded commands into code that can be understood by the DBMS
- *Example:*

```
//Program Segment E1:
0) loop = 1 ;
1) while (loop) {
2)   prompt("Enter a Social Security Number: ", ssn) ;
3)   EXEC SQL
4)     SELECT Fname, Minit, Lname, Address, Salary
5)     INTO :fname, :minit, :lname, :address, :salary
6)     FROM EMPLOYEE WHERE Ssn = :ssn ;
7)   if (SQLCODE == 0) printf(fname, minit, lname, address, salary)
8)   else printf("Social Security Number does not exist: ", ssn) ;
9)   prompt("More Social Security Numbers (enter 1 for Yes, 0 for No): ", loop) ;
10) }
```

Figure 10.2

Program segment E1,
a C program segment
with embedded SQL.